

KARL-GUSTAV PETERSON

Senior Backend Engineer - Node.js, Microservices, Cloud-Native Systems

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SUMMARY

Senior Backend Engineer with 10+ years of experience designing and scaling distributed systems using Node.js, Nest.js, and microservices architectures across AWS, Azure, and GCP. Specialized in building high-performance APIs, event-driven systems (Kafka, RabbitMQ), and real-time data pipelines with strong expertise in PostgreSQL, Redis, and distributed caching. Proven track record delivering production-grade backend platforms supporting 100K+ content systems, advanced search infrastructures, and secure payment integrations. Experienced in full stack and mobile development with React and React Native, bringing a strong end-to-end system perspective, while primarily focused on backend architecture, performance optimization, and cloud-native scalability.

CORE COMPETENCIES

- Backend Architecture
- Databases
- Real-Time Systems
- DevOps & CI/CD
- API Design
- Caching & Performance
- Authentication & Security
- AI Integration
- Distributed Systems & Event Streaming
- Search Platforms
- Cloud Platforms
- Observability

PROFESSIONAL EXPERIENCE

Senior Full-Stack & Backend Software Engineer

Solvelt | Warsaw, Poland | May 2022 - Mar 2026

- Led backend architecture for a 100K+ page medical platform, designing scalable **Node.js** microservices with **REST**, **GraphQL** and **gRPC** APIs to support high-volume content delivery and real-time workflows
- Designed event-driven systems using **Kafka** and **RabbitMQ**, enabling reliable asynchronous processing, real-time updates, and loosely coupled service communication
- Built and optimized **PostgreSQL schemas** and **indexing strategies**, improving query efficiency and supporting large-scale clinical data workloads
- Implemented **Redis**-based distributed caching and session management, reducing latency and improving system performance under high traffic
- Engineered secure backend services using **JWT**, **RBAC**, and audit logging, ensuring compliance-grade security for subscription and user data flows
- Developed payment systems using **Stripe API**, integrating **Apple Pay** and **Google Pay** with robust server-side validation and error handling
- Built unified search and indexing pipelines using **Elasticsearch** and **Algolia**, enabling fast, relevant queries across 40K+ clinical resources
- Containerized services with **Docker** and orchestrated via **Kubernetes (EKS)**, enabling horizontal scaling and zero-downtime deployments
- Integrated AI services (**OpenAI GPT-4o**, **Anthropic Claude**, and **GitHub Copilot**) into backend workflows for structured data extraction, summarization, and automation
- Implemented observability using **Grafana** and centralized logging, improving system monitoring, alerting, and operational reliability
- Collaborated with frontend and mobile teams using **Next.js** and **React Native** to deliver end-to-end, full-stack features

Senior Backend Software Engineer

Jelvix | Kharkiv, Ukraine | Aug 2020 - Apr 2022

- Architected scalable backend systems using **Node.js**, **Nest.js**, **Firebase Cloud Functions**, and **Azure Functions**, enabling modular microservices and serverless execution
- Designed secure, high-availability APIs with **REST**, **TypeScript**, and **Axios**, supporting real-time mobile and web applications
- Built real-time data synchronization pipelines using **Firestore**, **WebSockets**, and event-driven patterns for consistent cross-device state
- Implemented authentication and access control using **Firebase Authentication**, **RBAC**, and secure storage patterns
- Developed cloud-native storage and processing workflows using **Azure Blob Storage**, **Azure App Service**, and **Azure Static Web Apps**
- Engineered distributed notification systems using **Firebase Cloud Messaging**, **Expo Notifications**, and **Azure Notification Hubs**
- Integrated dynamic content pipelines via **Sanity CMS (GROQ)** and external APIs (**YouTube API**, **Mailchimp API**)
- Established observability and analytics using **Firebase Analytics** and **Azure Application Insights**, improving monitoring and product insights
- Automated backend workflows and integrations using **Azure Logic Apps**, **Cloud Scheduler**, and event-driven triggers
- Built and maintained CI/CD pipelines with **GitHub Actions**, **Azure DevOps Pipelines**, and **Expo EAS Build**, ensuring reliable releases

Backend Software Engineer

Opus Online | Tartu, Estonia | Nov 2017 – Jul 2020

- Designed and implemented backend services using **Node.js**, **REST**, and **GraphQL (Apollo Client)**, supporting scalable mobile-first architectures
- Built secure authentication systems using **JWT**, **OAuth**, and passwordless Magic Link, enabling flexible and secure user access flows
- Engineered real-time validation and transaction systems using **WebSockets**, ensuring low-latency and consistent data handling
- Optimized backend performance through efficient caching strategies, **Apollo Client caching**, and query optimization techniques
- Developed reusable backend-driven APIs and data layers to support cross-platform applications built with **React Native** and **Expo**
- Implemented scalable data-fetching and state synchronization strategies for high-performance mobile user experiences
- Contributed to mobile architecture using **React Navigation**, **Context API**, and modular design patterns
- Established testing and code quality standards using **Jest**, **React Native Testing Library**, **ESLint**, **Prettier**, and **Husky**
- Managed application releases and distribution via **App Store Connect** and **Google Play Console**

Backend Software Engineer

Brocoders | Tallinn, Estonia | Jul 2015 - Oct 2017

- Designed and deployed microservices-based backend systems using **Node.js** on **Google Cloud Platform**, leveraging **Cloud Run** and **Cloud Build** for scalable deployments
- Built robust API layers and integrated external services with fault-tolerant communication and retry strategies
- Developed modular frontend systems using **Next.js**, **TypeScript**, **Bootstrap**, and **Storybook**, aligned with backend service architecture
- Established CI/CD pipelines with automated testing using **Jest**, **Mocha**, and **React Testing Library**, ensuring consistent delivery quality
- Enforced clean architecture, **SOLID principles**, and TypeScript best practices through structured code reviews
- Implemented linting and code quality tooling using **ESLint**, **Prettier**, and **Husky**
- Delivered iterative features in Agile environments using cloud-native deployment workflows

TECHNICAL SKILLS

Backend Engineering & Architecture

- Node.js, Nest.js, TypeScript, JavaScript (ES6+)
- API Design: REST, GraphQL (Apollo Client/Server), gRPC
- Microservices architecture, modular monoliths, service decomposition
- Event-driven systems with Kafka, RabbitMQ
- Real-time systems using WebSockets
- Webhooks, background jobs, asynchronous processing pipelines

Data & Storage Systems

- Relational: PostgreSQL, MySQL, Oracle (schema design, indexing, query optimization)
- Caching & sessions: Redis (distributed caching, pub/sub, session stores)
- NoSQL: MongoDB, AWS DynamoDB, Firebase Firestore
- Data modeling, high-volume data processing, performance tuning

Search & Indexing

- Elasticsearch, Algolia
- Indexing pipelines, relevance tuning, query optimization
- Large-scale content search and real-time indexing workflows

Cloud & Distributed Infrastructure

- AWS (S3, CloudFront, IAM, Route 53, EKS)
- Azure (App Service, Static Web Apps, Functions, Blob Storage, Notification Hubs)
- Google Cloud Platform (Cloud Run, Cloud Build)
- Multi-region deployments, high availability, fault-tolerant systems

DevOps, CI/CD & Observability

- Containerization: Docker
- CI/CD: GitHub Actions, Azure DevOps, GCP Cloud Build, Expo EAS Build
- Monitoring & logging: Grafana, Azure Application Insights
- Orchestration: Kubernetes
- Release automation, infrastructure scaling, zero-downtime deployment strategies

Security, Authentication & Payments

- Authentication: JWT, OAuth 2.0, passwordless Magic Link
- Authorization: RBAC, secure access control patterns
- API security, server-side validation, audit logging
- Payments: Stripe API, Apple Pay, Google Pay

Frontend & Mobile (Supporting Strengths)

- Web: React.js, Next.js (SSR, SSG, ISR, App Router)
- UI: Tailwind CSS, Styled Components, Storybook, Atomic Design
- Mobile: React Native, Expo
- State management: Redux Toolkit, Context API
- Performance optimization: memoization, lazy loading, code splitting

Serverless & Integrations

- Firebase Cloud Functions, Azure Functions
- Firebase Authentication, Firestore
- Third-party integrations: Mailchimp API, YouTube API
- Workflow automation: Azure Logic Apps, Cloud Scheduler

AI & Data Intelligence

- OpenAI GPT-4o, Anthropic Claude, GitHub Copilot
- Semantic enrichment, content automation, AI-assisted workflows
- Structured output validation, prompt pipelines

Testing & Code Quality

- Testing: Jest, Mocha, React Testing Library, React Native Testing Library
- Code quality: ESLint, Prettier, Husky
- Unit, integration, and API testing strategies
- Code reviews, clean architecture, SOLID principles

EDUCATION

Bachelor's Degree in Computer Science

University of Tartu

Sep 2012 - Jun 2015

PROJECTS

Healthcare & Medical Education Platform

<https://GPnotebook.com>

Modernized and scaled GPnotebook, a 100K+ page clinical reference platform, by migrating from a legacy CMS to a composable architecture built with Next.js, React, TypeScript, Node.js, Sanity, Algolia, Stripe, and Vercel, enabling high-performance delivery, multi-region content management, CPD compliance tracking, subscription monetization, and unified web and mobile experiences for medical professionals across the UK, US, Australia, and Ireland.

InsurTech Full MVP Platform

<https://veygo.com>

Led development of a cross-platform mobile and web MVP using React Native, Expo, React.js, TypeScript, Node.js, Firebase, and Sanity CMS to validate a new InsurTech learning product. The platform enabled progress tracking, personalized content delivery, push notifications, analytics-driven user behavior insights, and secure customer data handling to support rapid idea validation and risk mitigation.